

HealthConnect Clinic — Week 6 Advanced Analytics & Decision Support Report

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Week 5 → Week 6 Transition

- 1. Main Week 5 output:** A cleaned dataset, 4 calculated KPIs (No-Show Rate by Reminder Status/Channel, Previous No-Show History, Distance to Clinic, and Booking Lead Time), an initial Power BI dashboard, and 5 business insights.
- 2. Most important result/component produced:** Booking Lead Time emerged as the strongest individual predictor of no-shows (correlation 0.272), with no-show rate more than doubling between short-notice bookings (24.84%) and appointments booked 22+ days out (56.94%).
- 3. Main issue or limitation discovered:** No single factor examined in Week 5 strongly predicts no-shows in isolation — even the strongest correlation (0.272) is modest. This suggests the true predictive power likely lies in how these factors *combine*, which Week 5's single-variable analysis could not capture.
- 4. Relevant track(s) for Week 6 activities:** Data Science — the relative strength of each Week 5 factor is directly relevant to their feature selection process for the no-show prediction model.
- 5. What I intend to improve, integrate or validate in Week 6:** Investigate combined/interaction effects between the strongest factors (e.g., does a patient with both high previous no-shows AND a long booking lead time show compounded risk beyond either factor alone), validate whether the Week 5 KPIs hold up under this deeper segmentation, and complete an actual integration exchange with the Data Science track — not just noting the connection, but exchanging concrete findings and documenting what changed as a result.

Combined Risk Analysis — Previous No-Show History × Booking Lead Time

Since Week 5's analysis examined each factor independently, and no single factor showed a strong correlation on its own (the strongest, Booking Lead Time, was only 0.272), Week 6 investigated whether combining the two strongest predictors reveals a compounding effect not visible in single-variable analysis.

Patients were segmented into four groups based on whether they had **any** prior no-show history and whether their appointment was booked with a **long lead time** (22+ days). The results show a clear compounding pattern:

- **Lowest risk** (no prior no-shows, short lead time): 29.94% no-show rate
- **Elevated risk from lead time alone** (no prior no-shows, long lead time): 51.70%
- **Elevated risk from history alone** (prior no-shows, short lead time): 39.87%
- **Highest risk** (prior no-shows AND long lead time): **64.09%**

This is a materially stronger and more actionable finding than either factor examined alone. A patient carrying both risk factors is more than twice as likely to no-show as a patient carrying neither — confirming that HealthConnect's most effective intervention strategy should target this **compound high-risk segment** specifically, rather than treating each risk factor as an independent, equally-weighted signal.

Combined Risk: Previous No-Shows × Booking Lead Time			
High Previous No-Show Risk	No	Yes	Total
No	29.94%	51.70%	43.51%
Yes	39.87%	64.09%	55.41%
Total	33.95%	56.94%	48.46%

Validating Week 5 KPIs Within the Combined High-Risk Segment

To confirm the Week 5 findings, remain meaningful once accounting for combined risk, reminder effectiveness and distance was re-examined specifically within the highest-risk group (patients with both prior no-show history and a long booking lead time).

- **Reminder effect holds, and matters more where it counts**
most: within the low-risk segment (no history, short lead time), reminders reduce no-show rate only marginally — from 31.16% to 29.49%, a modest ~1.7-point improvement. But within the **high-risk segment**, reminders show a larger absolute effect — from 67.39% (no reminder) down to 62.84% (reminder sent), a ~4.5-point improvement. This confirms reminders are not just generally helpful, but specifically most valuable for the patients who need them most.
- **Distance effect also holds within the high-risk group:** even after isolating the highest-risk segment, patients living 15km+ away still show a higher no-show rate (68.46%) than those within 15km (63.13%) — confirming distance contributes an additional, independent effect on top of history and lead time, rather than being fully explained away by those two factors alone.
- **Conclusion:** the Week 5 KPIs are validated as genuinely meaningful, not artifacts of overlapping variables — and this deeper analysis reveals a more precise, actionable insight: **reminder campaigns deliver the greatest impact when targeted at the combined high-risk segment (prior no-shows + long lead time), rather than applied uniformly across all patients.**

Cross-Track Integration

1. **Track collaborated with:** Data Science
2. **Project dependency:** The Data Science track's no-show prediction model (proposed in their Week 4 problem definition) requires validated, well-justified input features. Their Week 5 baseline model was built using individually proposed features without confirmation of which combinations carry the strongest predictive signal.
3. **Information/output received:** From the Data Science track's Week 4/5 documentation, the confirmed proposed target variable (appointment_outcome, specifically the No-Show class) and their initial candidate feature list, which treated previous_no_shows and booking_lead_days as independent inputs.
4. **Information/output provided:** The Week 6 combined-risk finding — that patients with both elevated previous no-show history AND a long booking lead time (22+ days) show a 64.09% no-show rate, more than double the rate for patients with neither factor (29.94%) — along with the specific recommendation to engineer a combined interaction feature (high_combined_risk) rather than treating these two variables independently.
5. **Integration activity completed:** Formalized this interaction effect into an explicit, implementable feature engineering specification, including the exact logic (previous_no_shows $\geq$ 1 AND booking_lead_days $\geq$ 22) and the quantified justification (the interaction shows more predictive signal than either variable alone).
6. **What changed as a result:** The Data Science track's feature set moves from treating these two variables as flat, independent inputs to including a purpose-built interaction feature directly informed by this week's analytical validation — a concrete change to their modeling approach grounded in evidence, not just a general observation.
7. **Evidence:** The Combined Risk Matrix (Previous No-Shows $\times$ Booking Lead Time) built in Power BI, showing the 4-cell breakdown (29.94% / 51.70% / 39.87% / 64.09%) that supports this recommendation, included in this report's Combined Risk Analysis section.

Combined Risk: Previous No-Shows × Booking Lead Time

High Previous No-Show Risk	No	Yes	Total
No	29.94%	51.70%	43.51%
Yes	39.87%	64.09%	55.41%
Total	33.95%	56.94%	48.46%

Business Recommendations

1. **Prioritize the combined high-risk segment for targeted intervention:** Patients with both prior no-show history AND a booking lead time of 22+ days show a 64.09% no-show rate — this group should receive the clinic's most intensive outreach (e.g., a confirmation call in addition to standard reminders), rather than spreading intervention effort evenly across all patients.
2. **Shift reminder strategy toward risk-based targeting rather than uniform application:** Since reminders show a larger absolute benefit within the high-risk segment (a ~4.5-point improvement) than the low-risk segment (~1.7 points), the clinic should prioritize reminder budget and staff time toward high-risk patients specifically, rather than treating all patients identically.
3. **Adopt SMS as the primary reminder channel:** SMS consistently outperformed WhatsApp and Email (45.75% vs. 48-50%), making it the most cost-effective channel to scale, especially when combined with the risk-targeting approach above.
4. **Consider a booking-lead-time policy review:** Since long lead times (22+ days) are the single strongest individual predictor of no-shows, the clinic could explore whether shorter default booking windows, or additional mid-point reminder touchpoints for long-lead appointments specifically, reduce this risk.
5. **Investigate distance-based support options for the small but high-risk 30km+ patient segment,** such as telehealth availability, given this group's elevated no-show rate (68.06%) despite its small size.

Data Limitations

The dataset remains fictional/synthetic, so findings are illustrative rather than directly deployable. The dataset does not capture *why* a patient no-showed, limiting root-cause certainty behind the patterns identified. The "5+ previous no-shows" and "30+km distance" segments remain based on small sample sizes and should be treated as directional. This week's interaction analysis, while revealing a strong combined effect, is based on only two factors — other unexamined combinations (e.g., involving distance or appointment type) may reveal further compounding effects not yet investigated.

Assumptions, Limitations, Risks and Dependencies (Week 6 Update)

- **Resolved:** The Week 4/5 limitation regarding missing distance_to_clinic_km and waiting_time_minutes values remains resolved (addressed via median imputation).
- **Remains unresolved:** The absence of a "reason for no-show" field continues to limit root-cause certainty.
- **New this week:** The interaction analysis is currently limited to two factors; a fuller multi-factor interaction model (potentially including distance) has not yet been tested and represents a new, identified area for Week 7.
- **New cross-track dependency:** The Data Science track's model now has an explicit dependency on this week's combined-risk finding for their feature engineering — if their model development reveals the interaction feature doesn't improve performance as expected, this should be fed back to the Data Analytics track for re-validation.
- **Potential impact:** If the combined-risk finding doesn't hold up under the Data Science track's model testing, the prioritized recommendations above (particularly #1 and #2) would need revisiting.
- **Recommended mitigation:** Maintain the feedback loop with Data Science into Week 7 to confirm the interaction feature's real predictive value before it becomes a settled operational recommendation.

Week 7 Analytical Testing Requirements

- Test whether adding distance as a third dimension to the combined-risk segmentation further refines or changes the highest-risk group identified this week.
- Validate whether the reminder-effectiveness pattern (stronger effect in high-risk segments) holds consistently across different appointment types and age groups, or whether it's concentrated in specific sub-groups.
- Cross-check the proposed `high_combined_risk` feature specification once the Data Science track has tested it in their model, to confirm it performs as expected before finalizing it as a recommended production feature.

Week 6 Project Summary — Data Analytics Track

1. **What I planned to accomplish:** Deepen and validate Week 5's individual-factor findings by investigating combined/interaction effects, rather than repeating the same EDA.
2. **What I completed:** Built and validated a combined-risk analysis crossing Previous No-Show History and Booking Lead Time; confirmed Week 5 KPIs remain meaningful within this deeper segmentation; updated the dashboard; completed a specific, evidence-based cross-track integration with Data Science.
3. **What I improved from Week 5:** Moved beyond single-variable KPIs to a genuine interaction analysis, revealing that combined risk (64.09%) is a substantially stronger and more actionable signal than any individual Week 5 factor.
4. **What I integrated:** A specific feature engineering recommendation (a combined interaction flag) for the Data Science track's predictive model, grounded in this week's validated findings.
5. **Which track(s) I collaborated with:** Data Science.
6. **What was exchanged:** I provided the combined-risk finding and a concrete feature engineering specification; I drew on their Week 4/5 documented target variable and feature list as the basis for the recommendation.
7. **What changed as a result:** Their candidate feature set now has a specific, evidence-justified interaction feature to include, rather than treating the two variables as flat, independent inputs.
8. **Key findings or development outcomes:** Combined risk (both factors present) yields a 64.09% no-show rate vs. 29.94% for neither — more than double. Reminders show a larger effect within the high-risk segment than the low-risk segment (67.39%→62.84% vs. 31.16%→29.49%). Distance continues to matter even within the high-risk segment.
9. **Major challenges:** Translating a technical interaction-effect finding into a clearly explainable, plain-language insight required extra care to ensure the reasoning was genuinely understood before building the visualization.
10. **Important decisions:** Simplified both variables into binary high/low risk flags (rather than keeping Week 5's 4-band structure) specifically to keep the combined-risk comparison clear in a 2×2 Matrix rather than a diluted 16-cell grid.
11. **Remaining issues:** Whether adding distance as a third risk dimension further refines the highest-risk segment has not yet been tested.

12. **My contribution to the overall HealthConnect project:** This week's work moves the project from "here are some factors that might matter" (Week 4-5) to "here is the specific, highest-priority patient segment to target, and here's the evidence-based feature the Data Science model should use to identify them" — a materially more decision-ready contribution.
13. **Proposed focus for Week 7:** Test whether distance adds meaningful refinement as a third risk dimension, and begin translating the full body of findings into final, prioritized business recommendations for HealthConnect leadership.